

PROJECT REPORT

IoT-Based Smart Plant Health & Soil Monitoring System

An ESP32-based system for real-time monitoring of soil moisture, temperature, humidity, and light intensity, with automated irrigation and IoT connectivity.

1. Abstract

Agriculture and plant cultivation require continuous monitoring of soil and environmental conditions. Conventional irrigation methods often depend on manual observation and can result in overwatering or underwatering.

This project presents an IoT-Based Smart Plant Health and Soil Monitoring System using an ESP32 microcontroller. The system measures soil moisture, temperature, humidity, and light intensity using multiple sensors.

The collected data is processed by the ESP32 to determine an environmental Plant Health Index. When soil moisture falls below a predefined threshold, the system automatically activates a water pump through a relay.

An OLED display provides local real-time information while Wi-Fi connectivity provides an IoT-ready platform for remote monitoring.

2. Introduction

Modern agriculture increasingly uses IoT technologies to improve resource efficiency and crop management.

Plants require suitable soil moisture, temperature, humidity, and light conditions for healthy growth. Monitoring these parameters manually is time-consuming and may result in delayed irrigation.

The proposed system automates environmental monitoring and irrigation using an ESP32.

3. Problem Statement

Traditional plant irrigation has several limitations:

- Manual monitoring is required.
- Water may be wasted.
- Soil moisture may not be checked regularly.
- Environmental conditions are difficult to monitor continuously.
- Farmers and gardeners may not have real-time information.

The proposed system addresses these problems using sensors, microcontroller-based processing, automatic irrigation, and IoT connectivity.

4. Objectives

The main objectives are:

- 1 Monitor soil moisture in real time.
- 2 Measure temperature and humidity.

- 3 Measure light intensity.
- 4 Display sensor readings locally.
- 5 Calculate an environmental Plant Health Index.
- 6 Automatically control irrigation.
- 7 Reduce unnecessary water usage.
- 8 Provide an IoT-ready monitoring system.

5. Proposed System

The system consists of four major sections:

5.1 Sensing Unit

The sensing unit consists of:

- Soil moisture sensor
- DHT22 temperature/humidity sensor
- BH1750 light sensor

These sensors collect information about the plant environment.

5.2 Control Unit

An ESP32 Development Board is used as the main controller. It receives sensor data, processes it, calculates the Plant Health Index, and controls the irrigation system.

5.3 Output Unit

The output section contains:

- OLED display
- RGB LED
- Relay
- Water pump

5.4 IoT Unit

The ESP32 connects to a Wi-Fi network and can transmit sensor data to an IoT dashboard or cloud platform.

6. Hardware Components

ESP32

The ESP32 is used because it provides:

- Wi-Fi
- Bluetooth
- Multiple GPIO pins
- ADC inputs
- I2C communication
- Sufficient processing capability

Soil Moisture Sensor

The sensor measures the moisture condition of the soil. Its analog output is connected to an ESP32 ADC pin.

DHT22

The DHT22 measures:

- Temperature
- Relative humidity

BH1750

The BH1750 measures ambient light intensity in lux.

OLED Display

The OLED displays live sensor readings and the Plant Health Index.

Relay

The relay acts as an interface between the ESP32 and the external water pump.

Water Pump

The pump supplies water to the plant when the soil becomes too dry.

7. Software Requirements

- Arduino IDE
- ESP32 Board Package
- DHT Library
- BH1750 Library
- Adafruit GFX Library
- Adafruit SSD1306 Library

8. System Working

The system operates as follows:

- 1 The ESP32 starts.
- 2 Sensors are initialized.
- 3 The ESP32 connects to Wi-Fi.
- 4 Soil moisture is measured.
- 5 Temperature and humidity are measured.
- 6 Light intensity is measured.
- 7 Sensor values are processed.
- 8 The Plant Health Index is calculated.
- 9 Values are displayed on the OLED.
- 10 The RGB LED indicates system status.
- 11 If soil moisture is below the threshold, the relay activates.
- 12 The water pump starts.
- 13 Once adequate moisture is restored, the pump stops.
- 14 Sensor data can be transmitted through Wi-Fi.

9. Plant Health Index

The project uses an experimental environmental suitability score.

The score is calculated using four parameters:

- Soil moisture
- Temperature
- Humidity
- Light

The maximum score is 100.

Example weighting:

Parameter	Weight
Soil Moisture	40%
Temperature	25%
Humidity	20%
Light	15%

The target ranges should be adjusted according to the plant species.

10. Automatic Irrigation

The irrigation system uses a predefined soil moisture threshold.

Example:

